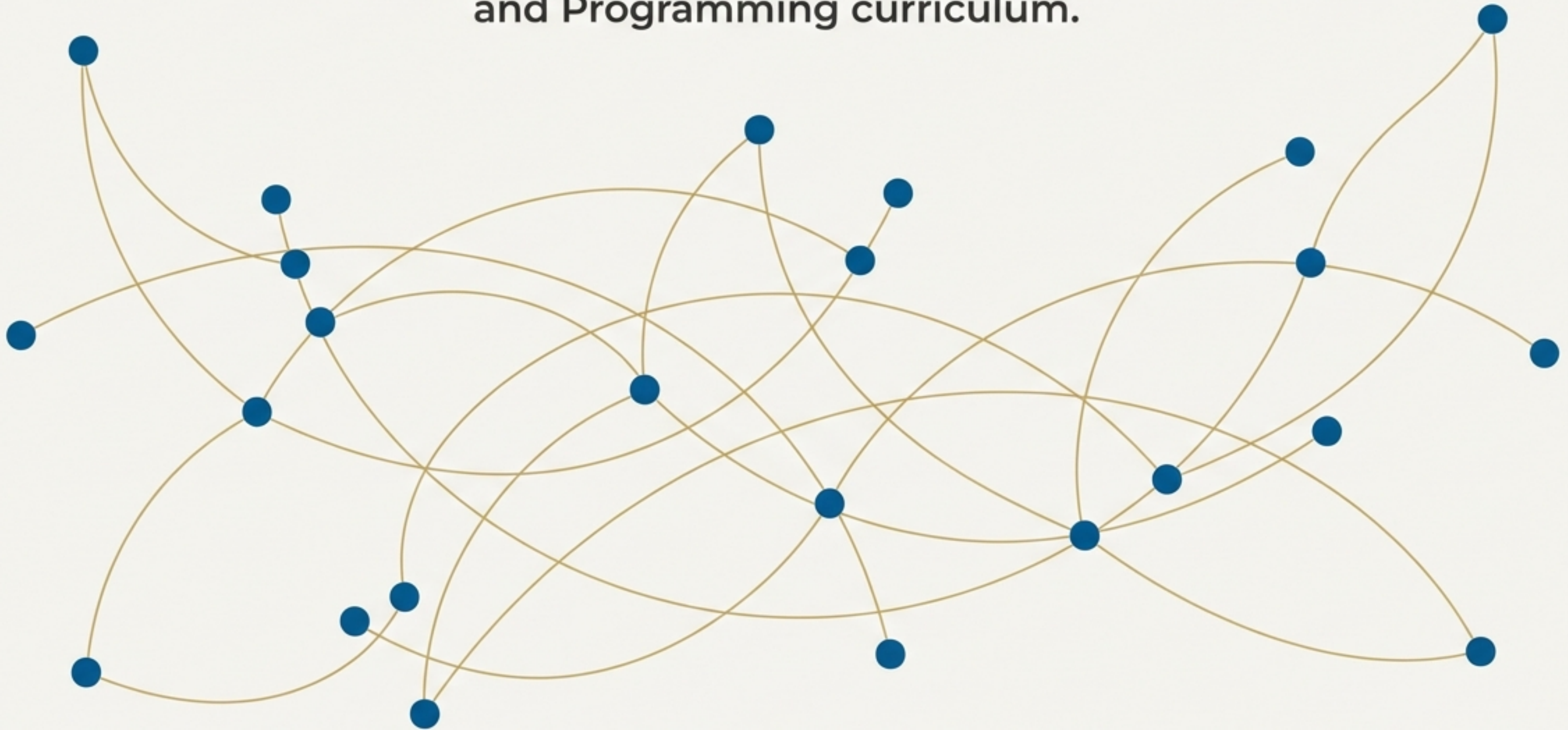


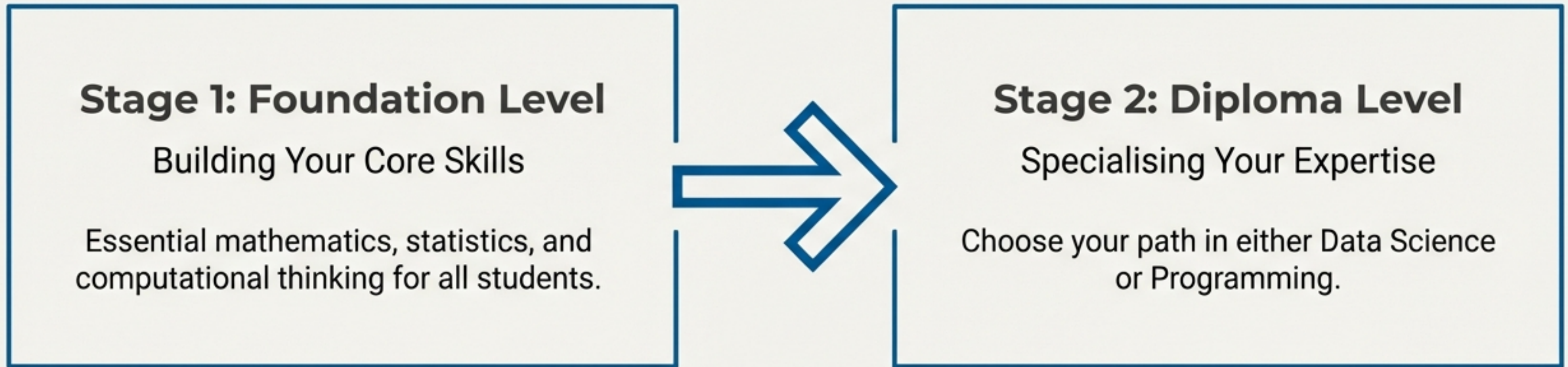
Charting Your Academic Voyage

Your definitive guide to navigating the Data Science
and Programming curriculum.



Your Programme is a Journey in Two Stages

The programme is designed to build your skills progressively. Every student begins with the Foundation Level before advancing to the Diploma Level, where you can specialise.



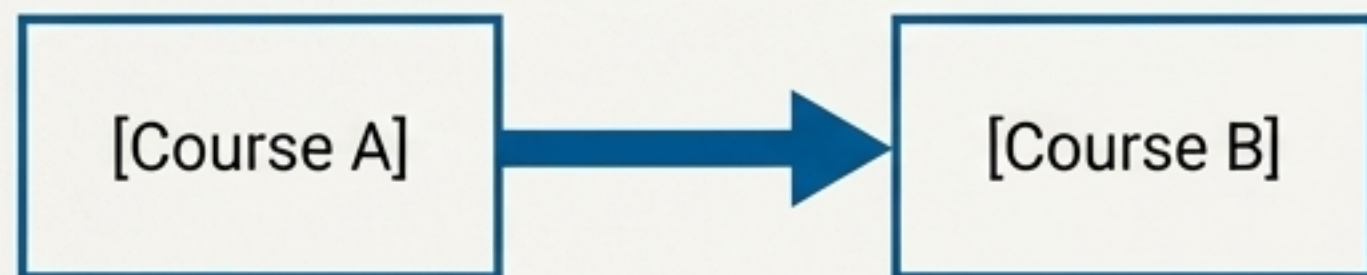
At each stage, you will choose a study pace that aligns with your personal and professional commitments.

Understanding Your Navigational Tools

Reading the Course Map

Prerequisite

A course you must complete *before* starting another.



Corequisite

A course you must take *at the same time* as another.



Choosing Your Travel Speed



The Accelerator Path

For full-time students or those with a strong background in maths and coding. An intense, fast-paced journey.



The Voyager Path

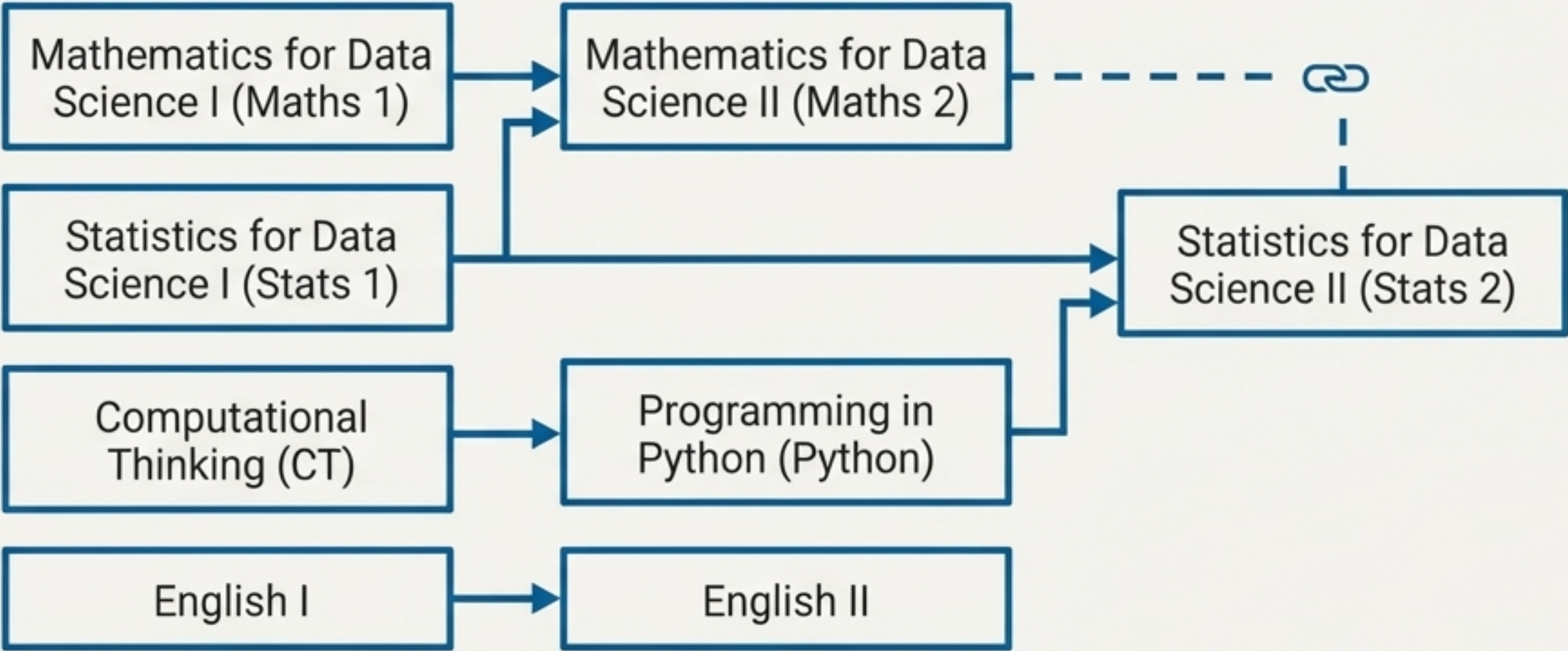
The recommended, balanced pace for most students, allowing for a deep understanding of the material while managing other commitments.



The Explorer Path

A comfortable, flexible pace designed for working professionals or those on a dual-degree track.

Stage 1: The Foundation Level Curriculum Map



Legend		
Course Name	Abbreviation	Code
Mathematics for Data Science I	Maths 1	BSMA1001
Statistics for Data Science I	Stats 1	BSMA1002
Computational Thinking	CT	BSCS1001
English I	English 1	BSHS1001
Mathematics for Data Science II	Maths 2	BSMA1003
Statistics for Data Science II	Stats 2	BSMA1004
Programming in Python	Python	BSCS1002
English II	English 2	BSHS1002

Foundation Level: Finding Your Pace



The Accelerator (2 Terms)

Recommended only for full-time learners or those strong in Maths/programming.

Term 1

- English 1
- Maths 1
- CT
- Stats 1

Term 2

- English 2
- Maths 2
- Python
- Stats 2



The Voyager (3 Terms)

A balanced pace for students with some foundational knowledge or those in another programme.

Term 1

- English 1
- Maths 1
- CT

Term 2

- Stats 1
- Maths 2
- English 2

Term 3

- Stats 2
- Python



The Explorer (4 Terms)

A comfortable and steady pace to ensure mastery of the fundamentals.

Term 1

- English 1
- Maths 1

Term 2

- Stats 1
- CT

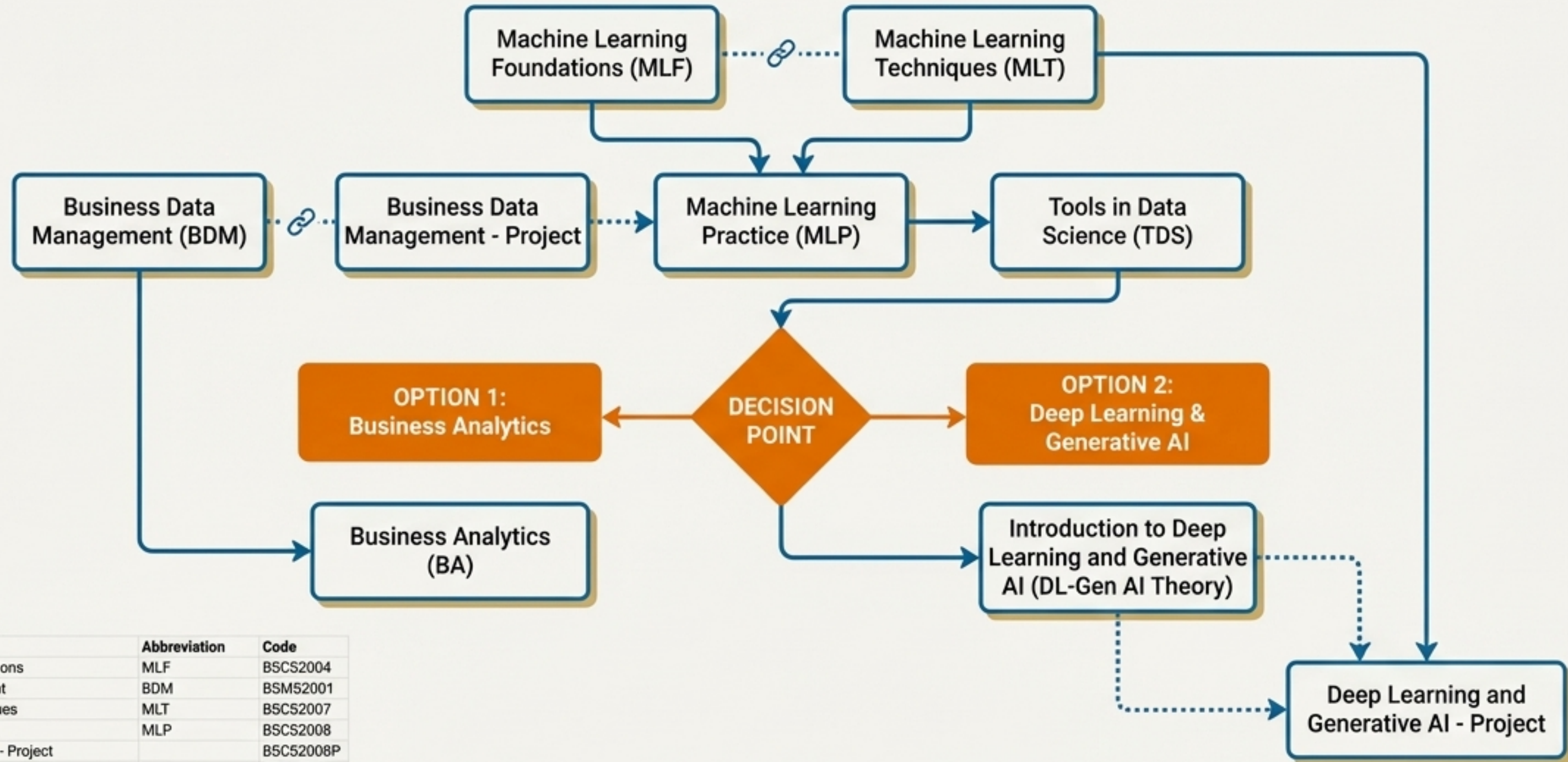
Term 3

- Maths 2
- English 2

Term 4

- Python
- Stats 2

Stage 2: The Diploma in Data Science Curriculum Map



Course Name	Abbreviation	Code
Machine Learning Foundations	MLF	BSCS2004
Business Data Management	BDM	BSM52001
Machine Learning Techniques	MLT	BSCS2007
Machine Learning Practice	MLP	BSCS2008
Machine Learning Practice - Project		BSCS2008P
Tools in Data Science	TDS	BSSE2002
Business Data Management - Project		B6M52001P
Business Analytics	BA	B5M52002
Introduction to Deep Learning and Generative AI	DL-Gen AI Theory	BSDA2001
Deep Learning and Generative AI - Project		BSDA2001P

Data Science Diploma: The Explorer and Voyager Paces



The Explorer Path (6-7 Terms)

A very comfortable path, highly recommended for working professionals and dual-degree students.

Table 1 (Business Analytics, 7-Term Plan)

Term 1	Term 2	Term 3	Term 4	Term 5	Term 6	Term 7
BDM Theory BDM Project SC	DBMS PDSA	MAD 1 Theory MLF	Mad 1 Project MLT	Mad 2 Theory MLP	Mad 2 Project BA MLP Project	Java TDS
9c	8c	8c	6c	8c	8c	7c

Table 2 (Deep Learning/Gen AI, 7-Term Plan)

Term 1	Term 2	Term 3	Term 4	Term 5	Term 6	Term 7
DBMS PDSA	MAD 1 Theory MLF	Mad 1 Project MLT SC	Mad 2 Theory MLP	MLP Project Mad 2 Project DL-Gen Ai Theory	DL-Gen Ai Project BDM	Java TDS
8c	8c	9c	8c	8c	6c	7c



The Voyager Path (5 Terms)

The correct speed for most standalone students, manageable if you are comfortable with Python and Maths.

Table 1 (Business Analytics, 5-Term Plan)

Term 1	Term 2	Term 3	Term 4	Term 5
DBMS PDSA MAD1 Theory	MLF MAD1 Project MAD2 Theory	BDM Theory MLT MAD2 Project	BDM Project MLP SC	BA Java TDS MLP Project
12c	10c	10c	9c	13c

Table 2 (Deep Learning/Gen AI, 5-Term Plan)

Term 1	Term 2	Term 3	Term 4	Term 5
DBMS PDSA MLF	BDM Theory MAD1 Theory MLT	MAD2 Theory MAD1 Project MLP	MAD2 Project SC MLP Project DL-Gen Ai Theory	TDS Java DL-Gen Ai Project
12c	12c	10c	11c	9c

Data Science Diploma: The Accelerator Pace



“Extremely hectic. Not recommended unless you are a full-time student and very strong in coding and maths.”

Table 1 (Business Analytics Focus, 4-Term Plan)

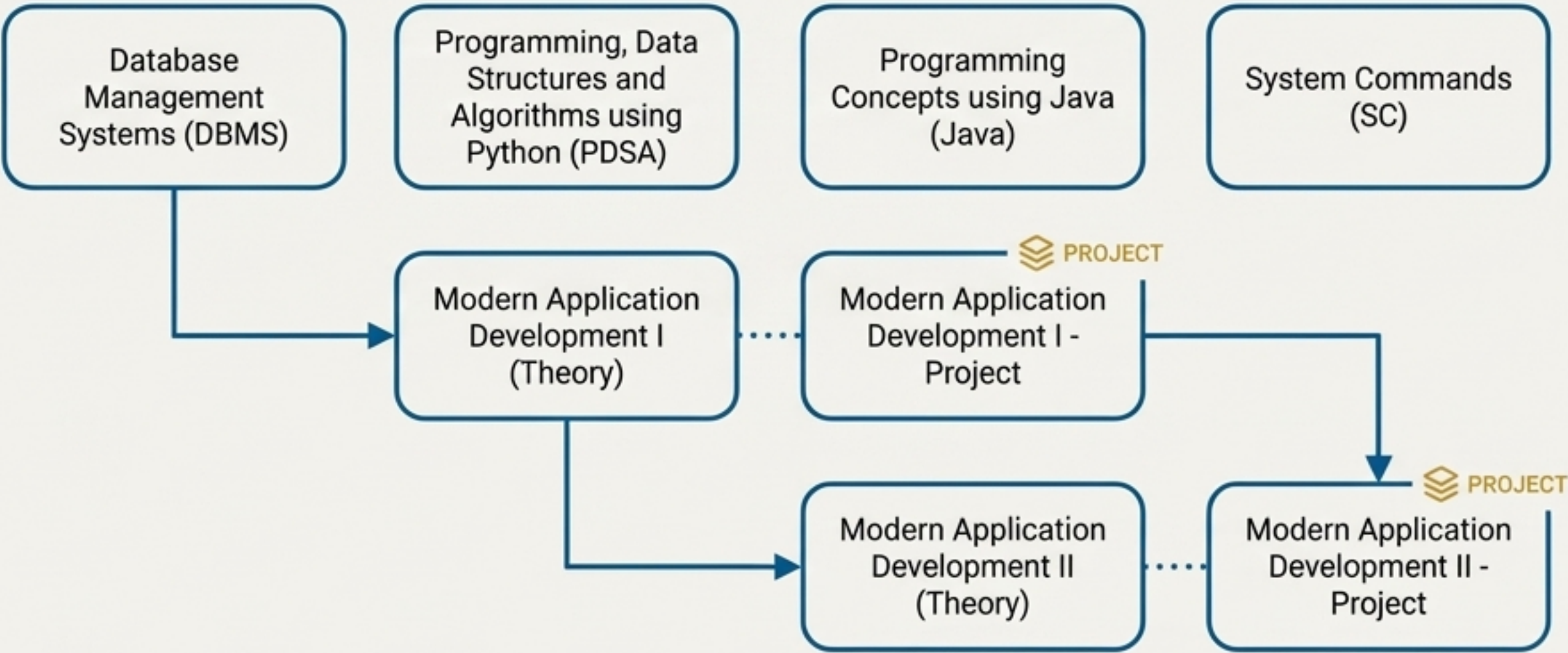
Term 1	Term 2	Term 3	Term 4
DBMS, BDM	Theory	BDM Project	Java
MAD1	MAD1 Project	BA	SC
PDSA	MAD2 Theory	MAD2 Project	TDS
MLF	MLT	MLP	MLP Project
16c	14c	12c	12c

Table 2 (Deep Learning/Gen AI Focus, 4-Term Plan)

Term 1	Term 2	Term 3	Term 4
DBMS, BDM	Theory	MAD2 Theory	MAD2 Project
PDSA	MAD1 Theory	MAD1 Project	TDS
MLT	MLP	MLP Project	Java
MLF	SC	DL-Gen Ai Theory	DL-Gen Ai Project
16c	15c	12c	11c

Programme completed in ~3 years.

Stage 2: The Diploma in Programming Curriculum Map



Code	Course Name
BSCS2001	Database Management Systems
BSCS2002	Programming, Data Structures and Algorithms using Python
BSCS2003	Modern Application Development I
BSCS2003P	Modern Application Development I - Project
BSCS2005	Programming Concepts using Java
BSCS2006	Modern Application Development II
BSCS2006P	Modern Application Development II - Project
BSSE2001	System Commands

Diploma in Programming: Finding Your Pace



The Accelerator (2 Terms)

For those on a fast track to development roles.

Term 1	Term 2
DBMS	App Dev 2 Th
App Dev 1 Th	App Dev 2 Proj
PDSA	SC
App Dev 1 proj	Java



The Voyager (3 Terms)

A steady, manageable pace for building programming skills.

Term 1	Term 2	Term 3
DBMS	App Dev 1 Project	App Dev 2 Proj
App Dev 1 Theory	App Dev 2 Theory	Java
	PDSA	SC



The Explorer (4 Terms)

A comfortable path for deep learning and practice.

Term 1	Term 2	Term 3	Term 4
App Dev 1	App Dev 2	PDSA	Java
DBMS	App Dev 1 project	App Dev 2 project	SC

Planning for a Single Diploma Exit in Data Science

These pathways are for students not completing the full degree programme.

Explorer (4 Terms)

**Extremely comfortable and relaxed pathway.*

Business Analytics (BA)

Term 1	Term 2	Term 3	Term 4
MLF	MLT	MLP	BA
BDM	BDM Project	TDS	MLP Project

Deep Learning/GenAI (DL)

Term 1	Term 2	Term 3	Term 4
MLF	MLP	MLP Project	TDS
MLT	BDM	DL GenAI Th	DL GenAI Proj

Voyager (3 Terms)

**Moderate pathway, if you are comfortable with Python and Maths.*

BA Option (BA)

Term 1	Term 2	Term 3
BDM	BA	TDS
BDM Project	MLT	MLP
MLF		MLP project

(DL Option)

Term 1	Term 2	Term 3
MLF	MLP	MLP proj
MLT	DL GenAI Th	DL GenAI Proj
	TDS	BDM

Accelerator (2 Terms)

****Not recommended unless you have a background and can handle this load.****

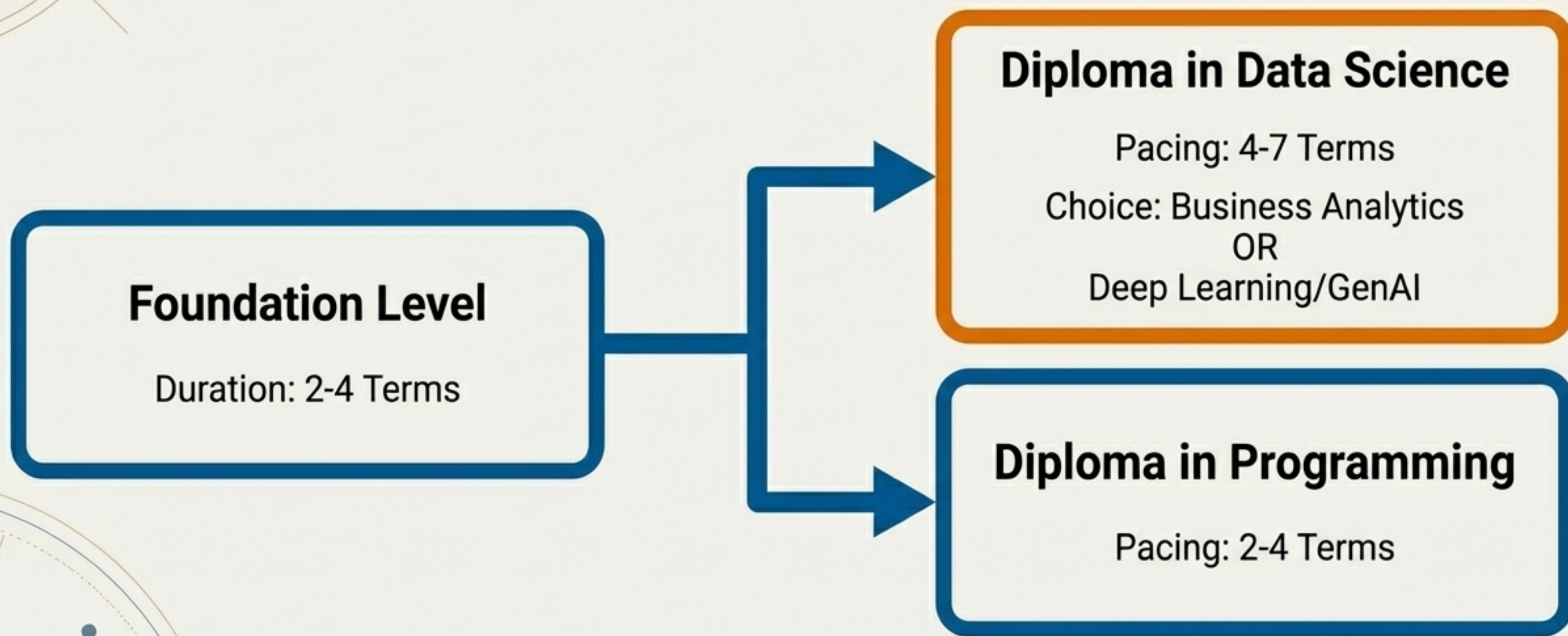
BA Option)

Term 1	Term 2
MLF	MLP
MLT	BA
BDM	TDS
BDM project	MLP proj

(DL Option)

Term 1	Term 2
MLF	MLP
MLT	DL GenAI Theory
BDM	MLP project
TDS	DL GenAI project

Visualising Your Complete Journey from Start to Finish



Your journey is customisable. Choose the pace and specialisation that's right for you.

Chart Your Course: Questions to Guide Your Decision

The right path is the one that fits your background, your schedule, and your ambitions. Before you decide, ask yourself:



1. What is my current expertise?

How strong are my Maths and Programming skills today? This will inform your Foundation Level pace.



2. What is my capacity?

How many hours per week can I realistically dedicate to my studies? This will help you choose between the Accelerator, Voyager, and Explorer paths.



3. What is my destination?

Do my career goals lean towards analytics and business strategy, or towards cutting-edge AI model development? This will guide your choice between the BA and DL/GenAI specialisations.



4. What is my context?

Am I a full-time student, a working professional, or pursuing a dual degree? This is the most critical factor in determining a sustainable pace for your journey.